Water balance

Purpose

The water balance files report the daily water budget of the land surface: precipitation and snow, evapotranspiration, surface runoff, lateral and tile flow, percolation, soil water and snowpack storage, and where the runoff goes. Every flux is a depth of water in mm over the area of the object, so objects of different size can be compared directly.

Four scopes are written, each switched on by a row in `print.prt`:

`print.prt` row	Files	One row per
`hru_wb`	`hru_wb_day.txt`, `hru_wb_mon.txt`, `hru_wb_yr.txt`, `hru_wb_aa.txt`	HRU
`hru-lte_wb`	`hru-lte_wb_day.txt`, `hru-lte_wb_mon.txt`, `hru-lte_wb_yr.txt`, `hru-lte_wb_aa.txt`	HRU-LTE
`lsunit_wb`	`lsunit_wb_day.txt`, `lsunit_wb_mon.txt`, `lsunit_wb_yr.txt`, `lsunit_wb_aa.txt`	landscape unit
`basin_wb`	`basin_wb_day.txt`, `basin_wb_mon.txt`, `basin_wb_yr.txt`, `basin_wb_aa.txt`	the whole basin (one row)

Each `y` in the `daily`, `monthly`, `yearly` or `avann` column of a row opens the matching file. With `csvout` set to `y`, a comma-separated `.csv` copy of each file is written as well.

`print.prt` also accepts a `region_wb` row, but 62.0.1 writes no region files, so the row has no effect. Routing units have no water balance file: the `ru` row writes routing unit hydrographs (`ru_day.txt` and so on; see Hydrograph outputs).

Format

- Line 1: the project name from `file.cio` and the SWAT+ version banner.
- Line 2: column headers.
- Line 3: units.
- Lines 4 and on: one row per object per print step.

The first seven columns are the same in every landscape output file:

#	Column	Description
1	`jday`	Day of the year of the last day in the print step (for example 31 for a January row of a monthly file).
2	`mon`	Month.
3	`day`	Day of the month.
4	`yr`	Calendar year.
5	`unit`	Sequence number of the object (HRU number, landscape unit number). The basin file always shows 1.

#	Column	Description
6	gis_id	GIS identifier of the object from its connection file. The basin file always shows 1.
7	name	Object name. The basin file shows the project name.

The remaining columns, in file order:

#	Column	Units	Description
8	precip	mm	Precipitation, rain and snow, before canopy interception.
9	snofall	mm	Precipitation that fell as snow.
10	snomlt	mm	Snowmelt.
11	surq_gen	mm	Surface runoff that leaves the HRU on the day, after the surface runoff lag.
12	latq	mm	Lateral flow out of the soil profile.
13	wateryld	mm	Water yield: surq_gen + latq + qtile.
14	perc	mm	Percolation out of the bottom of the soil profile.
15	et	mm	Actual evapotranspiration: ecanopy + eplant + esoil.
16	ecanopy	mm	Evaporation of water intercepted by the canopy.
17	eplant	mm	Plant transpiration.
18	esoil	mm	Soil evaporation, including evaporation of ponded water.
19	surq_cont	mm	Surface runoff generated on the HRU on the day, before the surface runoff lag. For an HRU with a wetland, this is the wetland outflow.
20	cn	-	Curve number used for the day.
21	sw_init	mm	Soil water content at the start of the print step.
22	sw_final	mm	Soil water content at the end of the print step.
23	sw_ave	mm	Average soil water content over the print step.
24	sw_300	mm	Soil water content of the top 300 mm of soil.
25	sno_init	mm	Snow water content at the start of the print step.
26	sno_final	mm	Snow water content at the end of the print step.
27	snopack	mm	Average snow water content over the print step.
28	pet	mm	Potential evapotranspiration.
29	qtile	mm	Tile drain flow.
30	irr	mm	Irrigation water applied.
31	surq_runon	mm	Surface runoff received from upslope objects.
32	latq_runon	mm	Lateral flow received from upslope objects.
33	overbank	mm	Overbank flood water received from channels.
34	surq_cha	mm	Surface runoff sent to channels.
35	surq_res	mm	Surface runoff sent to reservoirs.
36	surq_ls	mm	Surface runoff sent to other landscape objects.
37	latq_cha	mm	Lateral flow sent to channels.
38	latq_res	mm	Lateral flow sent to reservoirs.
39	latq_ls	mm	Lateral flow sent to other landscape objects.
40	gwsoilq	mm	Groundwater moved into the soil profile when the water table rises into it (gwflow only).
41	satex	mm	Saturation excess flow from a high water table (gwflow only).
42	satex_chan	mm	Saturation excess flow that reaches a channel (gwflow only).
43	sw_change	mm	Change in soil water content over the day.
44	lagsurf	mm	Surface runoff held back by the lag and still on its way to the channel.
45	laglatq	mm	Lateral flow held back by the lag and still on its way to the channel.
46	lagsatex	mm	Saturation excess flow held back by the lag and still on its way to the channel.
47	wet_evap	mm	Evaporation from the wetland water surface.
48	wet_oflo	mm	Wetland outflow (spill).

#	Column	Units	Description
49	wet_stor	mm	Water stored in the wetland.
50	plant_cov	text	Plant community of the HRU's land use, from landuse.lum.
51	mgt_ops	text	Management schedule of the HRU's land use, from landuse.lum.

The columns surq_cha to latq_ls split the HRU's generated surface runoff and lateral flow by receiving object, using the fractions in the HRU's connection file. They are filled for HRUs only and stay zero in HRU-LTE rows.

Trailing columns

The header line lists plant_cov and mgt_ops in every file, but only the hru_wb files write them. The other files end after wet_stor, except basin_wb_aa.txt, which ends with the soft calibration label and the adjustment value: Original Simulation and 0.000 when no soft calibration is run. The label is two words, so a tool that splits on spaces finds one more value than the header has names.

Aggregation

- Daily rows are written for days inside the print.prt daily window (day_start of yrc_start to day_end of yrc_end), every interval days.
- Monthly, yearly and average annual rows start after the nyskip warm-up years.
- Fluxes (every column except the storages listed below) are summed over the month or year. The average annual value is the sum over the printed years divided by the number of printed years.
- cn, sw_ave, sw_300, snopack and wet_stor are averaged over the days in the step. In the average annual file they are averaged over all printed days.
- sw_init and sno_init give the value at the start of the step, and sw_final and sno_final the value at the end. In *_aa.txt, sw_final and sno_final are the values on the last day of the simulation.

Landscape unit and basin rows are area-weighted averages of their HRUs and HRU-LTEs, using the area fractions in ls_unit.ele.

Example

The header line of hru_wb_aa.txt as 62.0.1 writes it, split here over several lines:

```
 jday   mon   day    yr    unit  gis_id  name           precip     snofall      snomlt    surq_gen
       latq    wateryld        perc          et     ecanopy      eplant       esoil   surq_cont
         cn     sw_init    sw_final      sw_ave      sw_300    sno_init   sno_final     snopack
        pet       qtile         irr  surq_runon  latq_runon    overbank    surq_cha    surq_res
    surq_ls    latq_cha    latq_res     latq_ls     gwsoilq       satex  satex_chan   sw_change
    lagsurf     laglatq    lagsatex    wet_evap    wet_oflo    wet_stor   plant_cov      mgt_ops
```

To check an average annual HRU row, compare precip + irr + surq_runon + latq_runon with et + wateryld + perc plus the change in soil water and snow storage. A large gap usually points to wetland storage, gwflow exchange or water held in the runoff lag.

The reference outputs in refdata/Ames_sub1 of the SWAT+ repository were written by revision 61.0.1, and their header has no plant_cov and mgt_ops names.

Related files

- print.prt: switches each file on or off and sets the print window.
- Output naming convention: every output file and the print.prt row that controls it.
- Channel and sediment outputs: where the water leaving the landscape goes next.
- Post-processing helpers: reading these files with Python.

Sources

- File opening and header writes: output_landscape_init.f90

- Record, header and units types, and the add and average operators: `output_waterbal`, `output_waterbal_header` and `output_waterbal_header_units` in `output_landscape_module.f90`
- Daily values: `hru_control.f90`, with the runoff lag in `sq_surfst.f90` and wetland outflow in `wetland_control.f90`
- Writers: `hru_output.f90`, `hru_lte_output.f90`, `lsu_output.f90`, `basin_output.f90`
- `print.prt` reader: `basin_print_codes_read.f90`
- Print window and step flags: `time_control.f90`

Column	Fortran variable
`surq_gen`	`qday` (after `sq_surfst`)
`wateryld`	`qdr(j)`
`perc`	`sepbtm(j)`
`et`	`etday`
`ecanopy`	`canev`
`eplant`	`ep_day`
`esoil`	`es_day`
`surq_cont`	`surfq(j)`
`cn`	`cnday(j)`
`sw_ave`	`output_waterbal%sw`
`pet`	`pet_day`
`irr`	`irrig(j)%applied`
`surq_runon`	`ls_overq`
`latq_runon`	`latqrunon`
`overbank`	`hru(j)%wet_obank_in`
`gwsoilq`	`output_waterbal%gwsoil`, from `gwsoilq(j)`
`sw_change`	`output_waterbal%delsw`
`lagsurf`, `laglatq`, `lagsatex`	`surf_bs(1,j)`, `bss(1,j)`, `bss_ex(1,j)`
`wet_oflo`	`output_waterbal%wet_out`
`wet_stor`	`wet(j)%flo / (10 * area_ha)`